

# Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-07-28.jpg":

$$L = \lim_{x \rightarrow 0^+} \frac{\sin(x^2) \int_0^{\sqrt{x}} \sin(t^2) dt}{\sin(\sqrt{x}) \int_0^{x^2} \sin(\sqrt{t}) dt}$$

---

## Detailed Solution

To evaluate the limit  $L$  as  $x \rightarrow 0^+$ , we determine the leading-order asymptotic behavior (or Taylor series expansion around 0) for each individual factor and integral in both the numerator and the denominator.

### Step 1: Asymptotic Behavior of the Trigonometric Factors

Recall the standard limit  $\lim_{u \rightarrow 0} \frac{\sin(u)}{u} = 1$ , which implies that  $\sin(u) \sim u$  as  $u \rightarrow 0$ . We apply this approximation to the trigonometric terms outside the integrals:

$$\begin{aligned}\sin(x^2) &= x^2 + \mathcal{O}(x^6) \implies \sin(x^2) \sim x^2 \quad \text{as } x \rightarrow 0^+ \\ \sin(\sqrt{x}) &= \sqrt{x} + \mathcal{O}(x^{3/2}) \implies \sin(\sqrt{x}) \sim \sqrt{x} \quad \text{as } x \rightarrow 0^+\end{aligned}$$

### Step 2: Asymptotic Expansion of the Definite Integrals

Next, we approximate the definite integrals by expanding their integrands for small values of the integration variable  $t$ .

**Numerator Integral:** As  $x \rightarrow 0^+$ , the upper limit of integration  $\sqrt{x} \rightarrow 0$ . Within the range of integration  $t \in [0, \sqrt{x}]$ , we use the expansion  $\sin(t^2) = t^2 + \mathcal{O}(t^6)$ :

$$\int_0^{\sqrt{x}} \sin(t^2) dt \sim \int_0^{\sqrt{x}} t^2 dt = \left[ \frac{t^3}{3} \right]_0^{\sqrt{x}} = \frac{(\sqrt{x})^3}{3} = \frac{1}{3} x^{3/2}$$

**Denominator Integral:** Similarly, as  $x \rightarrow 0^+$ , the upper limit  $x^2 \rightarrow 0$ . For  $t \in [0, x^2]$ , we use the expansion  $\sin(\sqrt{t}) = t^{1/2} + \mathcal{O}(t^{3/2})$ :

$$\int_0^{x^2} \sin(\sqrt{t}) dt \sim \int_0^{x^2} t^{1/2} dt = \left[ \frac{2}{3} t^{3/2} \right]_0^{x^2} = \frac{2}{3} (x^2)^{3/2} = \frac{2}{3} x^3$$

### Step 3: Evaluating the Final Limit

We now substitute the leading-order approximations back into the original expression for  $L$ :

$$\begin{aligned} L &= \lim_{x \rightarrow 0^+} \frac{\sin(x^2) \int_0^{\sqrt{x}} \sin(t^2) dt}{\sin(\sqrt{x}) \int_0^{x^2} \sin(\sqrt{t}) dt} \\ &= \lim_{x \rightarrow 0^+} \frac{x^2 \cdot \left(\frac{1}{3}x^{3/2}\right)}{\sqrt{x} \cdot \left(\frac{2}{3}x^3\right)} \\ &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{3}x^{7/2}}{\frac{2}{3}x^{7/2}} \end{aligned}$$

Since the powers of  $x$  in the numerator and denominator are identical ( $x^{7/2}$ ), they cancel out completely:

$$L = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

**Final Answer:**

$$\frac{1}{2}$$